

Paper 1 - Set 14 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	A	Graphs of functions
2	C	Ratio, proportion and percentages
3	F	Number, units and estimation
4	B	Circles and circle theorems
5	B	Differentiation
6	B	Number, units and estimation
7	C	Polynomials, factor and remainder theorems
8	F	Polynomials, factor and remainder theorems
9	A	Quadratic functions
10	D	Circles and circle theorems
11	A	Integration
12	D	Functions and transformations
13	D	Exponentials and logarithms
14	D	Trigonometric functions and equations
15	F	Simultaneous equations and inequalities
16	E	Quadratic functions
17	D	Pythagoras, mensuration and 3D
18	A	Functions and transformations
19	C	Triangles, radians and sectors
20	A	Trigonometric functions and equations

Worked solutions

Question 1 GRA-10 · Graphs of functions · spec MM8.6

A polynomial of degree 5 with real coefficients has how many real roots at least, and at most?

Answer: A

An odd-degree polynomial tends to opposite infinities at each end, so it crosses the axis at least once; a degree-5 polynomial has at most 5 roots.

Question 2 RAT-13 · Ratio, proportion and percentages · spec M3.8

The value of a car falls by 20% each year. After how many complete years is it first worth less than half its original value?

Answer: C

$0.8^3 = 0.512 > 0.5$, $0.8^4 = 0.4096 < 0.5$. So 4 years.

Question 3 NUM-14 · Number, units and estimation · spec M2.12

Simplify $\sqrt{48} \times \sqrt{27} \div \sqrt{12}$.

Answer: F

$\sqrt{48} = 4\sqrt{3}$, $\sqrt{27} = 3\sqrt{3}$, $\sqrt{12} = 2\sqrt{3}$. So $\frac{4\sqrt{3} \times 3\sqrt{3}}{2\sqrt{3}} = \frac{36}{2\sqrt{3}} = \frac{18}{\sqrt{3}} = 6\sqrt{3}$.

Question 4 CIR-10 · Circles and circle theorems · spec MM3.3

TA is a tangent to a circle at A , and AB is a chord. C is a point on the circle on the opposite side of AB from T , with angle $ACB = 65^\circ$. Find angle TAB .

Answer: B

Alternate segment theorem: the angle between tangent and chord equals the angle in the alternate segment, so $TAB = 65^\circ$.

Question 5 DIF-06 · Differentiation · spec MM6.1

The displacement s metres of a particle at time t seconds is $s = t^3 - 6t^2 + 9t$. At what times is the particle momentarily at rest?

Answer: B

Velocity $= \frac{ds}{dt} = 3t^2 - 12t + 9 = 3(t-1)(t-3) = 0$ at $t = 1$ and $t = 3$.

Question 6 NUM-13 · Number, units and estimation · spec M2.7

Which of the following is equal to $\frac{8^4 \times 2^{-5}}{4^3}$?

Answer: B

$\frac{2^{12} \cdot 2^{-5}}{2^6} = 2^1 = 2$.

Question 7 POL-07 · Polynomials, factor and remainder theorems · spec MM1.6

Find the quotient when $2x^3 - 3x^2 + 4x - 5$ is divided by $(x - 1)$.

Answer: C

$$2x^3 - 3x^2 + 4x - 5 = (x - 1)(2x^2 - x + 3) - 2: \text{ check the expansion}$$

$$2x^3 - x^2 + 3x - 2x^2 + x - 3 = 2x^3 - 3x^2 + 4x - 3, \text{ then subtract 2 for the remainder.}$$

Question 8 POL-01 · Polynomials, factor and remainder theorems · spec MM1.6

When $x^3 - 4x^2 + kx + 6$ is divided by $(x - 3)$ the remainder is -6 . Find k .

Answer: F

$$\text{Remainder Theorem: } 27 - 36 + 3k + 6 = -6 \Rightarrow 3k - 3 = -6 \Rightarrow k = -1.$$

Question 9 QUA-03 · Quadratic functions · spec MM1.3

The quadratic $y = ax^2 + bx + c$ has its vertex at $(2, -3)$ and passes through $(0, 5)$. Find $a + b + c$.

Answer: A

$$y = a(x - 2)^2 - 3; \text{ at } x = 0: 4a - 3 = 5, \text{ so } a = 2. \text{ Then } y = 2(x - 2)^2 - 3 = 2x^2 - 8x + 5, \text{ and}$$

$$a + b + c = 2 - 8 + 5 = -1 \text{ (which is just } y(1)).$$

Question 10 CIR-04 · Circles and circle theorems · spec MM3.2

The points $(1, 7)$ and $(5, -1)$ are the ends of a diameter of a circle. Find the equation of the circle.

Answer: D

$$\text{Centre is the midpoint } (3, 3); \text{ radius}^2 = (3 - 1)^2 + (3 - 7)^2 = 4 + 16 = 20.$$

Question 11 INT-10 · Integration · spec MM7.2

Find $\int \left(x + \frac{1}{x}\right)^2 dx$.

Answer: A

$$\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + x^{-2}, \text{ which integrates to } \frac{x^3}{3} + 2x - x^{-1} + c. \text{ (Option E is a common error: you cannot integrate a power of a bracket that way.)}$$

Question 12 FUN-16 · Functions and transformations · spec MM1.7

The function f is defined by $f(x) = |x - 2| + |x + 2|$ for all real x . What is the minimum value of $f(x)$, and for how many values of x is it attained?

Answer: D

$$\text{For } -2 \leq x \leq 2: f(x) = (2 - x) + (x + 2) = 4. \text{ Outside this interval } f(x) > 4. \text{ So the minimum 4 is attained on the whole interval } [-2, 2].$$

Question 13 LOG-19 · Exponentials and logarithms · spec MM5.3

A quantity halves every 5 hours. Which expression gives the number of hours for it to fall to $\frac{1}{10}$ of its initial value?

Answer: D

$$\left(\frac{1}{2}\right)^{t/5} = \frac{1}{10}. \text{ Taking logs to base 10: } -\frac{t}{5}\log_{10} 2 = -1, \text{ so } t = \frac{5}{\log_{10} 2} \text{ (about 16.6 hours).}$$

Question 14 TRG-15 · Trigonometric functions and equations · spec MM4.4

The graph of $y = \sin x$ (with x in radians) crosses the line $y = x$ at how many points?

Answer: D

$|\sin x| < |x|$ for all $x \neq 0$, so the only intersection is at the origin.

Question 15 SIM-15 · Simultaneous equations and inequalities · spec MM1.5

The solution set of $ax^2 + bx + c < 0$ (with $a > 0$) is $-2 < x < 5$. What is the solution set of $ax^2 - bx + c < 0$?

Answer: F

Replacing x by $-x$ reflects the graph in the y -axis, so the roots -2 and 5 become 2 and -5 : $-5 < x < 2$.

Question 16 QUA-04 · Quadratic functions · spec MM1.3

The roots of $x^2 - 5x + 2 = 0$ are α and β . Find the value of $\alpha^2 + \beta^2$.

Answer: E

$\alpha + \beta = 5$ and $\alpha\beta = 2$. So $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 25 - 4 = 21$.

Question 17 MEN-10 · Pythagoras, mensuration and 3D · spec M5.7

A square-based pyramid has base side 6 and all slant edges of length 9. Find its height.

Answer: D

Half the base diagonal is $3\sqrt{2}$, so $h^2 = 81 - 18 = 63$.

Question 18 FUN-07 · Functions and transformations · spec MM8.2

The function f satisfies $f(x) = x^2$ for all x . Which of the following transformations maps $y = f(x)$ onto $y = 4x^2 - 8x + 4$?

Answer: A

$4x^2 - 8x + 4 = 4(x - 1)^2$, i.e. $y = 4f(x - 1)$: translate 1 right, then stretch vertically by 4. (Equivalently $y = f(2x - 2) = f(2(x - 1))$: stretch parallel to the x -axis by factor $\frac{1}{2}$ then translate 1 right - not offered.)

Question 19 TRI-16 · Triangles, radians and sectors · spec MM4.2

Two circles of radius 2 have their centres a distance 2 apart. Find the area of the region common to both circles.

Answer: C

The common chord subtends $\frac{2\pi}{3}$ at each centre (equilateral triangles of side 2). The overlap is two segments:
 $2 \times \frac{1}{2} \cdot 4 \left(\frac{2\pi}{3} - \sin \frac{2\pi}{3} \right) = 4 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) = \frac{8\pi}{3} - 2\sqrt{3}$.

Question 20 TRG-10 · Trigonometric functions and equations · spec MM4.6

How many solutions does the equation $\sin x = \frac{x}{10}$ have?

Answer: A

Solutions need $|x| \leq 10$. Both sides are odd functions, so count $x > 0$ and double, then add $x = 0$. On $(0, \pi)$ the curve starts above the line (since $\sin x \approx x$) and ends below it: one crossing. On $(\pi, 2\pi)$, $\sin x < 0$: none. On $(2\pi, 3\pi)$ the line is between 0.63 and 0.94 while $\sin x$ rises to 1: two crossings ($3\pi \approx 9.42 < 10$). Beyond $4\pi > 10$ the line exceeds 1. So 3 positive, 3 negative and 0: seven solutions.